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Inclusive design, all astronauts at the heart of design - Future-Proofing Human Spaceflight for All

Saira O. Williams^{a*}, Aditi Sant^b, Bouchra Harnoufi^c, Helen Haile^d

^a *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^b *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^c *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^d *University of Nottingham, United Kingdom, University Park, NG7 2RD, Nottingham, U. K*

* Corresponding Author

Abstract

As human spaceflight progresses toward long-duration missions and permanent extraterrestrial settlements, the design of space habitats must evolve to accommodate a diverse astronaut workforce. Traditional spacecraft and habitats have been developed with limited accessibility considerations, often requiring adaptations rather than integrating inclusive design from the outset. However, with initiatives such as ESA's Parastronaut Feasibility Project and the growing recognition of the importance of diversity in crew selection, rethinking habitat design is essential to ensure that all astronauts can live and work effectively in space.

This paper explores key challenges and opportunities in designing future-proof space habitats, prioritising inclusivity. By examining how accessibility can be embedded into fundamental aspects of habitat development, we highlight the need for adaptable environments, human-centred interfaces, and redefined safety protocols that support astronauts of varying abilities. Addressing these issues will broaden astronaut selection criteria and enhance overall mission safety, efficiency, and long-term sustainability. Ensuring universal access to space habitats is not only a matter of equity—it is a strategic imperative for the future of space exploration. As we push beyond Earth, inclusive design must be at the heart of the systems and structures supporting human presence beyond our planet.

Keywords: Inclusive Design; Astronaut Selection; Accessibility; Space Habitats; Human Spaceflight; Policy and Innovation

Acronyms/Abbreviations

AI – Artificial Intelligence
ESA – European Space Agency
EVA – Extravehicular Activity
ISS – International Space Station
NASA – National Aeronautics and Space Administration
PLSS – Portable Life Support System
SDGs – Sustainable Development Goals
UNOOSA – United Nations Office for Outer Space Affairs

1. Introduction

Human spaceflight is entering a new phase, moving from short missions in low Earth orbit to long-duration expeditions and future settlements. These ambitions demand spacecraft and habitats that support astronauts of different backgrounds and abilities. Historically, design and selection relied on narrow physiological standards, with accessibility considered only after the fact. Such approaches risk limiting crew diversity and mission resilience [1].

1.1 Policy Context

Institutions are beginning to address these limitations. ESA's Parastronaut Feasibility Project introduced a new astronaut category, focusing initially on candidates with lower-limb impairments such as below-the-knee amputations or significant leg-length differences [2]. Similarly, research commissioned by NASA and Potomac Institute for Policy Studies recommends revising medical standards and adapting spacecraft interiors to accommodate astronauts with physical disabilities such as vestibular disorders, hearing impairments, vision loss conditions, and others that do not compromise cognitive performance or operational capability [3]. At the international level, the UNOOSA links disability inclusion in space to the Sustainable Development Goals, framing accessibility as both equity and innovation [4].

1.2 Participation and lived experience

Practical demonstrations reinforce these policy shifts. Mission: AstroAccess has flown disabled scientists, veterans, and artists on parabolic flights, testing tactile, auditory, and mobility-related adaptations in microgravity [5]. This year, Blue Origin hosted an AstroAccess workshop where deaf and hard-of-hearing ambassadors completed New Shepard's training and proposed accessibility best practices, marking a milestone toward commercial participation [5].

This research paper positions inclusive design as a strategic imperative for human spaceflight. By examining policy frameworks, lived experiences, and

cross-sector impacts, it argues that universal access to spacecraft and habitats is essential for safety, sustainability, and future mission success.

2. Rethinking the Design Baseline

It has been 60 years since humans started designing space habitats. Since the USSR launched the first human space station on April 19, 1971, around 34 space stations have been sent into space for research and military reconnaissance. Mir was the first space station for humans' long-term presence in space. It took about ten years to assemble the individual specialised modules after its launch in 1986, including the recent ones, the ISS (International Space Station) and the Tiangong program by China, designed for three crew members, on September 29, 2011[24]. Human Space flight is entering a new era, with the growing interest in human space flight for further settlement on the Moon and Mars. Recent innovations in spacesuit design and the ESA para-astronaut selection process for the future diverse astronaut workforce highlight the importance of a new thinking approach to shape the future of human space flight.

2.1 Innovation in Spacesuit Design

The SpaceX spacesuit for the Crew Dragon spacecraft, unlike traditional bulky spacesuits with a Portable Life Support System (PLSS), the SpaceX spacesuit for the Crew Dragon spacecraft relies on the Dragon Crew spacecraft's life support system, making it a lightweight suit as well as offering flexibility, adjustability, and comfort for astronauts of different body types and fitted to the astronauts' individual body types, including for women[19].



Figure 2.1 SpaceX Spacesuit [27]

Similarly, the Axiom Extravehicular Mobility Unit (AxEMU) was developed through collaboration between Axiom and NASA. It is designed to accommodate a wide range of body types and genders. This will offer improved mobility and performance during spacewalks. The suit's design approach is user-centred, enabling more flexibility and easy use so that the astronaut can plug in when they sit down, and the suit takes care of things from there, reducing the mental and physical strain during the operation.



Figure 2.2 Axiom Extravehicular Mobility Unit (AxEMU) [23]

These collaborations and innovations between NASA, Axiom and SpaceX have shown that space exploration is heading into a new era where human-centric approaches are prioritised, rather than purely operational engineering constraints. Further facilitating the rethinking of the future inclusive habitat design by centring the inhabitant and human factor-derived design, moving beyond the traditional engineering Human Space flight engineering problems[22][23].

2.2 Limitations of Current Baselines

The past missions, such as MIR and ISS, have successfully enabled highly trained astronauts to travel and operate in microgravity environments since the start of human space travel, despite their design limitations. The early design approach for human space travel addresses the five fundamental problems: artificial gravity, environmental constraints such as radiation, microgravity and temperature, efficient radiation protection, sustainable food, growth strategy, and societal value[32].

However, the Psychological and physiological needs of the inhabitants were secondary, often assuming a healthy, physically and psychologically fully capable crew

2.2.1 Physical Accessibility

Human space travel presents unique technological and environmental challenges unlike aviation flights. Early missions, including the Apollo and some current selections, relied on crews with military backgrounds who were accustomed to a restrictive and uncomfortable environment. This influenced the early habitat design to prioritise mission functionality over human-centric aspects, such as physical comfort or inclusivity.

This is evident in NASA's suit and NASA's Exploration Extravehicular Mobility Unit (xEMU) designs. For example, the Apollo mission suits and xEMU designs have benefited from custom tailoring and some adjustability; some minor sizing issues were corrected in flight. However, the challenges remain regarding women's suits, as the early designs were based on male anthropometry. Still have some difficulties regarding flexibility, adjustability, and comfort for astronauts of different body types and fitted to the astronauts' individual body types, including for women. This limits Physical mobility, particularly reducing astronaut mobility and comfort, even as modern suits such as the Space Crew Dragon and the suggested AxEMU will improve adaptability and user comfort [20][21].

Physical Accessibility encompasses mobility, flexibility, usability within the habitat, considering diverse body types. Legacy design assumptions inherently limit inclusivity, highlighting the need for habitats and equipment that accommodate varied physical capabilities. Applying principles such as the Universal Space Theory in Architecture, which emphasises openness, adaptation, and flexibility for space inhabitants, can guide the design of habitats that support physical mobility. Future approaches should create adaptable and comfortable environments for all astronauts, enabling safe and efficient operation regardless of individual physical differences.[29]

2.2.2 Cognitive Accessibility

The Traditional habitat and mission designs assume that astronauts will arrive in good physical and mental condition, ready to perform tasks immediately upon arrival. [30] However, microgravity and long-duration missions can impact cognitive performance. Changes in brain structure due to microgravity may affect motor and mental functions, leading to fatigue, while isolation,

confinement, and environmental stressors further challenge cognitive resilience.[31]

Cognitive accessibility involves interface design, control systems, and management of information overload. Inclusive habitats require intuitive, multisensory interfaces and redundancy to ensure that astronauts with diverse cognitive profiles can operate safely and effectively under operational stress [30][31][32].

2.2.3 Emergency and Safety Protocols and Accessibility

Emergency and safety procedures often assume uniform physical and cognitive capabilities. Critical alerts, life-support panels, and emergency protocols may be inaccessible for astronauts with reduced mobility, strength, or sensory impairments. Designing inclusive emergency systems requires multiple approaches including adaptable equipment, orientation-agnostic modules, and training scenarios that reflect a diversity of abilities[31].[32].

3. Inclusive Design Principles & Testing

Ensuring accessibility in future space habitats requires more than retrofitting existing systems. It calls for a shift in how we approach design, testing, and evaluation from the outset. Inclusion must be structurally embedded into each stage of engineering—from requirements definition to system validation. This section outlines how universal design principles can be systematically integrated through inclusive testing methodologies, multisensory interfaces, and performance evaluation frameworks that reflect a diversity of users.

3.1 Embedding Diversity into Testing Methodologies

Current spacecraft systems have historically been designed and tested using narrow user profiles typically based on military or test pilot standards. This has excluded many individuals with sensory, cognitive, or physical disabilities from the design loop. To move toward inclusive spaceflight, testing protocols must evolve to reflect the growing diversity of astronaut candidates [1][6].

A recent example presented at the 74th International Astronautical Congress (IAC 2023) proposed an experimental protocol to evaluate procedure comprehension for individuals with hearing impairments, in simulated space mission conditions [2]. This protocol compared various formats—text-only, image-based, text+image, and silent video—for

delivering operational instructions under realistic constraints. This design offers a replicable methodology for inclusive testing without requiring immediate human spaceflight access.

Such frameworks demonstrate how accessibility considerations can be integrated upstream in the validation process, ensuring that future systems can accommodate a broader range of operational profiles.

3.2 Multisensory Interfaces and Cognitive Accessibility

Accessibility is not limited to physical parameters; it also encompasses sensory and cognitive usability. Future space habitats must account for a wide spectrum of interaction needs. This includes designing interfaces that support multisensory input/output, such as:

- Redundant cues (visual, auditory, tactile, haptic),
- Consistent use of symbols and colour-coding,
- Voice or gesture control as alternatives to manual input,
- Clear layout hierarchies to minimise cognitive overload [7].

Designers should involve diverse user profiles in early testing—such as neurodivergent individuals or people with sensory impairments—to ensure these elements are not only functional but intuitive under operational stress. This aligns with established best practices in universal design [8], but must be adapted for extreme environments such as microgravity and isolation.

3.3 Inclusive Performance Metrics

Evaluating the usability of space systems traditionally centres on task efficiency, speed, and error rate. However, inclusive design requires expanding these metrics to include:

- System adaptability to individual needs (e.g., adjustable displays, flexible input modes),
- User comprehension and confidence, especially under stress,
- Robustness of interaction in non-standard conditions (gloved hands, limited visibility, degraded communication).

The NASA Human Integration Design Handbook already highlights the importance of system flexibility and human-centred design [9]. But to support inclusive missions, future versions should also define performance thresholds that accommodate variable capabilities rather than expecting conformity to a single operational profile.

3.4 Toward Standardised Inclusive Testing

To ensure sustainable and scalable inclusion in human spaceflight, accessibility testing must become part of industry and agency standards, not remain isolated research efforts. This involves:

- Mandating accessibility evaluation in procurement requirements,
- Including diverse users in design reviews and simulation campaigns,
- Harmonising protocols across agencies to create unified guidelines.

ESA's Parastronaut Feasibility Project is a concrete step in this direction, showing how candidate evaluation and system design can adapt to accommodate mobility impairments without compromising mission safety [1].

Ultimately, inclusive testing strengthens system resilience—not just for astronauts with disabilities, but for all crew members operating in unpredictable and extreme conditions.

4. Adaptive, Personalised, and Modular Technologies

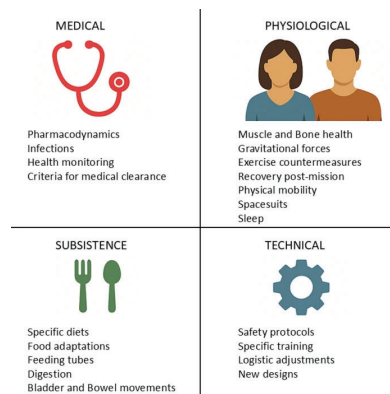


Fig. 3. Challenges in Human Space Research - Preliminary considerations for accessible space missions for all. npj Microgravity 11, 29 (2025).

Long-duration missions require habitats that evolve with crew needs rather than one-size-fits-all designs. Future cabins should support multimodal interfaces, combining voice control, gesture recognition, touch input, and even eye-tracking so that any astronaut regardless of ability or context can operate systems naturally [11]. Studies of augmented-reality controls show that integrating speech, gaze, and hand gestures yields more intuitive, accessible interaction, enhancing attention, spatial commands, and hands-free operation [11]. Designing with such redundancy means that a

spoken command, a button push, or a simple eye-movement/flick can achieve the same result.

4.1 Modularity and Reconfigurable Habitats

Habitat furniture and fixtures should be built of interchangeable modules or soft robots so they can shift configuration on demand. For instance, walls, workbenches, and storage units could be rearranged, retracted or extended to suit different tasks or users. Recent analyses recommend 'modular and flexible' habitat layouts (i.e. movable handrails, foldaway surfaces) to accommodate varying body sizes, mobility aids, or sensory needs [12]. A workstation might automatically adjust its height and orientation if a tall crew member floats in to use it or collapse into a compact shape when not needed. This ensures every crew member (*wheelchair users to far-sighted technicians*) can easily reach controls and stowage.

4.2 Personalisation of Environmental Conditions

Crew comfort is boosted when individuals can tweak their surroundings. Habitats should allow personalized 'ambience profiles', i.e. adjustable lighting, temperature, humidity, and noise insulation for each person. Research on confined habitats underscores that giving occupants individual control over lighting and climate greatly improves comfort and mood [13]. In practice, modern spacecraft already exploit tunable LEDs: the ISS uses adjustable color-temperature lights that crew members can set (warm for sleep, cool for work) to counter isolation stress and regulate circadian rhythms [14]. Similarly, interfaces should remember personal display and language preferences. These customizations help prevent overstimulation or fatigue, especially for neurodiverse astronauts or those with sensory sensitivities.

4.4 Designing for Lifespan Diversity

Designs must anticipate that crew members' bodies and missions change. The same habitat might later host older explorers, injured astronauts, or even children. Thus, designers should avoid fixed, narrow passageways or one-size equipment. Instead, planning for adjustable harnesses, versatile restraint systems, and adaptable interfaces will help serve a wide age range and ability mix [15]. For example, seat restraints that work both under gravity and in freefall (via quick-release harnesses) are being prototyped by NASA. In effect, every new generation of settlers can be accommodated by simply reconfiguring the existing infrastructure.

4.5 AI-Driven Adaptation and Smart Systems

On-board intelligence can learn and predict crew needs. AI agents could automatically optimize the

habitat environment (for example, gently increasing lighting before a crewmember with poor vision needs to read, or filtering ambient noise around someone sensitive to sound). ‘AI-driven adaptive interfaces’ are an important future direction for such environments [11]. Smart textiles and robotics are also envisioned, e.g., self-adjusting exercise equipment or smart fabrics that monitor vital signs and adjust padding or support. Importantly, designing these intelligent systems with accessibility in mind (i.e. voice-operated diagnostics for an injured astronaut) will create overall more robust habitat controls that benefit the entire crew [12].

5. Accessibility in Emergencies and Operations

Emergencies in space demand split-second responses, so critical alerts and tools must be usable by anyone on board. Inclusive emergency systems save lives.

5.1 Multimodal Alerts and Instructions

Alarms and warnings should never rely on just one sense. NASA’s crew-interface standards mandate both visual indicators and audible alarms for any emergency or caution [15]. In practice, this means flashing lights paired with horns or speech announcements. Building on this, tactile alerts (vibrating suits/wristbands) can further ensure nobody misses a warning, especially if an astronaut is visually or hearing impaired. Research confirms that ‘multimodal’ (visual + auditory + haptic) signaling significantly improves situational awareness in hostile environments like EVA, reducing response times and errors [16]. All emergency instructions (escape routes, checklist steps,) should likewise be available in redundant formats: text, audio narration, and simple symbols, so they are clear to diverse users.

5.2 Emergency Equipment Accessibility

Life-support panels, medical kits, fire extinguishers, and EVA suits must be operable by crew with limited dexterity or mobility. This could mean oversized buttons or grips that work with gloves or prosthetic limbs, and color-coding of critical valves and handles. NASA and ESA studies highlight that space suits and tools should be designed with accommodations (for example, space suits with alternate hand controls for prosthetic fingers to allow easy donning and operation) [2]. The ESA Parastronaut project specifically investigates how to build suits and gear (with adaptive harnesses, modified gloves, etc.) so that any astronaut can, say, secure their oxygen tank or wield a power drill under duress [2].

Wearable robotic augmentation can further extend accessibility and safety. Zhao et al. present the AstroLimbs system, a set of reconfigurable robotic limbs designed to support astronauts during

extravehicular activities (EVAs). These limbs provide up to 6 degrees of freedom for assistance in gripping, stabilizing, or manipulating tools, even when mobility or strength is limited. Such systems could benefit crew with reduced dexterity, prosthetic users, or fatigued astronauts in emergencies.

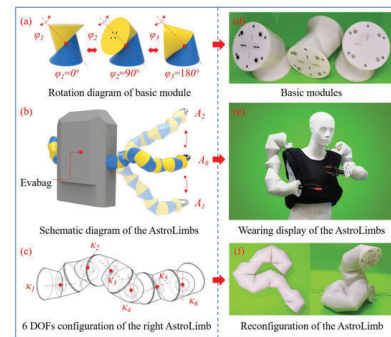


Fig. 4. Wearable robotic limbs for astronaut extravehicular activities assistance. Adapted from Zhao et al. [17]

5.3 Safe Zones and Orientation-Agnostic Modules

Crucial shelters (for example fire-safe compartments, medical bays) should be accessible from multiple directions and via tactile guides. For instance, modules can be designed with handrails and footholds on all surfaces so crew can crawl or float toward them from any orientation. Tactile floor or wall markers (like grooved strips) can ‘lead’ unconscious or visually impaired crew to safety. In microgravity, even the notion of floor and ceiling is fluid; hence habitat layouts should be orientation-agnostic. Every chamber should have multiple entrances/exits and storage that can serve as impromptu seats or stretchers. While few direct studies exist, accessible architecture and space analogs strongly suggest that such redundancy dramatically improves egress safety [8].

5.4 Inclusive Crisis Simulation and Training

Safety drills must include diverse participants. Training scenarios should simulate low-visibility or equipment-failure conditions for crew members using assistive devices or prosthetics. As one expert review points out, mixed-ability training helps crews learn to assist each other (for example, practicing how to help an immobilized astronaut don an emergency harness) [9]. Diverse drills reveal hidden flaws in emergency plans and raise awareness. For example, practicing a capsule evacuation with a blindfolded or one-armed crewmate can test whether signaling procedures or tie-down points are truly universal.

5.5 System Robustness through Inclusion

In practice, designing for accessibility often makes systems better for everyone. Redundant, multimodal communications (visual, audio, vibration) not only aid a disabled user, but also provide backup in case one system fails. Studies of design-for-all in aerospace note that initially ‘accommodating’ special needs drives innovation: the resulting interfaces and layouts are more ergonomic and resilient [2][9]. In short, an emergency system that is simple enough for the least mobile astronaut to operate will also be more intuitive for a fully able one, reducing errors under stress. By building this redundancy and universality from the start, we improve mission safety and overall crew performance.

6. Policy, Participation & Cross-Sector Impact

Inclusive design in human spaceflight is shaped by intersecting forces of policy, lived participation, and cross-sector collaboration. Progress in these domains demonstrates that broadening astronaut eligibility is not only a matter of equity but also a pathway to safer, more resilient, and more innovative missions.

6.1 Policy Drivers

Policy frameworks are redefining the boundaries of astronaut eligibility. ESA’s Parastronaut Feasibility Project set a precedent by opening recruitment to candidates with lower-limb impairments such as below-the-knee amputations or significant leg-length differences [1]. While narrow in scope, it marked an institutional shift towards inclusion. In the United States, the NASA–Potomac Institute for Policy Studies has recommended adapting medical standards and spacecraft interiors for a wider spectrum of conditions, including prosthetic use, vestibular disorders, hearing impairments, and vision loss [2]. At the global level, UNOOSA has positioned disability inclusion in space as part of the United Nations Disability Inclusion Strategy and the Sustainable Development Goals, urging member states to embed accessibility in future treaties and exploration programs [4].

6.2 Participation in Practice

Policy initiatives gain credibility when supported by practice. Mission: AstroAccess has conducted a series of parabolic flights involving blind, low-vision, deaf, hard-of-hearing, and mobility-impaired participants [5]. These flights evaluated tactile wayfinding, auditory and visual alarms, captioning systems, and transfer techniques in microgravity. In 2025, collaboration with Blue Origin advanced this work further: deaf and hard-of-hearing ambassadors completed New Shepard

training and developed accessibility guidelines for suborbital flight [5].

Historic precedents also underscore the value of inclusion. In the late 1950s, NASA partnered with the Gallaudet Eleven, a group of deaf men whose vestibular systems were nonfunctional to study balance, motion sickness, and orientation in extreme environments. Their unique physiology allowed them to perform in conditions where hearing astronauts became incapacitated, providing crucial data for early human spaceflight [18]. This example illustrates that individuals with disabilities have long offered operational advantages in space research.

6.3 Cross-Sector Impact and implications for Future Missions

Accessibility measures tested in spaceflight yield benefits far beyond astronautics. In healthcare, tactile and multimodal alarms developed for spacecraft can improve safety in hospitals and emergency response. In technology, haptic and auditory interfaces originally designed for blind astronauts inform universal design in consumer electronics [5]. Education initiatives linked to inclusive astronaut programs inspire underrepresented groups to pursue STEM careers, strengthening the future workforce. At the policy level, cross-sector dialogues led by UNOOSA illustrate how space can drive innovation in inclusive governance [4].

Long-duration missions to the Moon and Mars will expose crews to unpredictable conditions. Embedding inclusivity into selection, training, and design not only broadens participation but also strengthens mission resilience. Accessibility adaptations create redundancy, diversify crew skills, and improve overall safety, ensuring that future exploration represents humanity in both its diversity and capability.

6. Conclusions

Humanity’s next steps into deep space demand more than advanced propulsion or robust life-support systems, they require rethinking how we design for the human experience itself. Inclusive design ensures that space systems are not narrowly tailored to a small subset of individuals, but rather engineered to serve astronauts of all abilities, ages, and backgrounds.

Evidence across policy, lived experience, and technical research confirms that accessibility adaptations increase mission safety, reduce operational risks, and expand the talent pool for astronaut selection. Initiatives such as ESA’s Parastronaut Feasibility Project and Mission:

AstroAccess demonstrate that inclusivity is not only possible but beneficial. Technical measures, multisensory interfaces, modular habitats, AI-driven adaptive environments, and inclusive emergency systems—strengthen resilience for all astronauts, not only those with disabilities.

Beyond spaceflight, accessibility innovations developed for extreme environments influence healthcare, education, and human-computer interaction on Earth. Designing for diversity in orbit catalyzes solutions with universal value.

For upcoming missions to the Moon, Mars, and beyond, inclusive design must be embedded at the earliest stages of planning. Only by placing all astronauts at the heart of design can we secure a resilient, equitable, and sustainable future for human space exploration.

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References

- [1] European Space Agency (ESA), “Apply now to become an ESA astronaut – including the Parastronaut Feasibility Project,” Press Release, 2021. Available at: https://www.esa.int/Newsroom/Press_Releases/Apply_now_to_become_an_ESA_astronaut(accessed 28.09.2025).
- [2] Potomac Institute for Policy Studies; NASA Office of the Chief Health and Medical Officer (OCHMO), Parastronaut Feasibility – Foundational Research Study, Washington, DC, 2022. Available at: https://www.potomacinstitute.org/images/Publications/PDF/PIPS_OCHMO_Parastronaut_Feasibility_FullReport.pdf (accessed 28.09.2025).
- [3] NASA Technical Reports Server, Expanding Access to Spaceflight: A Preliminary Assessment of Accommodations for Specific Disabilities on Spacecraft Design and Operations, 2024. Available at: <https://ntrs.nasa.gov/citations/20240008257> (accessed 28.09.2025).
- [4] United Nations Office for Outer Space Affairs (UNOOSA), “Space4Persons with Disabilities,” 2023. Available at: <https://www.unoosa.org/oosa/en/ourwork/space4personswithdisabilities/space4persons-with-disabilities.html> (accessed 28.09.2025).
- [5] Mission: AstroAccess, “AstroAccess and Blue Origin Collaborate to Advance Accessibility in Spaceflight,” 2025. Available at: <https://astroaccess.org/astroaccess-and-blue-origin-collaborate-to-advance-accessibility-in-spaceflight/> (accessed 28.09.2025).
- [6] SGAC Diversity & Gender Equality Project Group, “DIVINAS: Astronauts with Disabilities – Research and Experiment on the Disability Inclusion in the Human Space Program,” in Proc. 74th Int. Astronaut. Congr. (IAC), Baku, Azerbaijan, 2023, IAC-23-B3.3.8.
- [7] M.F. Story, “Maximizing usability: the principles of universal design,” *Assist. Technol.* 10 (1) (1998) 4–12.
- [8] E. Steinfeld, J. Maisel, *Universal Design: Creating Inclusive Environments*, John Wiley & Sons, Hoboken, 2012.
- [9] NASA, *Human Integration Design Handbook (HIDH)*, NASA/SP-2010-3407, Rev. 1, 2014.
- [10] I. Di Giulio, R. Anderton, N. Caplin, et al., “Preliminary considerations for accessible space missions for all,” *npj Microgravity* 11 (29) (2025). <https://doi.org/10.1038/s41526-025-00494-9>
- [11] S. Zaheer, “Advancing AR Interfaces: Integrating Gesture, Voice, and Eye-Tracking for Enhanced Interaction,” *Int. J. Leading Research Publication* 5 (6) (2024). <https://doi.org/10.5281/zenodo.15259169>
- [12] K. Amini, M. Moradi, B. Vossoughi, E.D. Janabadi, “Space-technological and architectural methodology and process towards design of long-term habitats for scientific human missions on Mars,” *MethodsX* 11 (2023) 102270. <https://doi.org/10.1016/j.mex.2023.102270>
- [13] H. He, “Home far away from home: Spacecraft interior design for human well-being,” *Intersect: The Stanford Journal of Science, Technology, and Society* 16 (2) (2023). Available at: <https://ojs.stanford.edu/ojs/index.php/intersect/article/view/2480> (accessed 28.09.2025).

- [14] A. Jiang, X. Yao, S. Westland, C. Hemingray, B. Foing, J. Lin, "The Effect of Correlated Colour Temperature on Physiological, Emotional and Subjective Satisfaction in the Hygiene Area of a Space Station," *Int. J. Environ. Res. Public Health* 19 (15) (2022) 9090. <https://doi.org/10.3390/ijerph19159090>
- [15] NASA, 10-0 Crew Interfaces (Vol. 2), 18 Feb. 2025. Available at: <https://www.nasa.gov/reference/10-0-crew-interfaces-vol-1-2/> (accessed 28.09.2025).
- [16] E.M. Wenzel, M. Godfroy-Cooper, "Advanced multimodal solutions for information presentation," NASA/TM-20210017507, NASA Ames Research Center, 2021. Available at: <https://ntrs.nasa.gov/citations/20210017507> (accessed 28.09.2025).
- [17] S. Zhao, J. Zhao, D. Sui, T. Wang, T. Zheng, C. Zhao, Y. Zhu, "Modular Robotic Limbs for Astronaut Activities Assistance," *Sensors* 21 (18) (2021) 6305. <https://doi.org/10.3390/s21186305>
- [18] NASA, "How 11 Deaf Men Helped Shape NASA's Human Spaceflight Program," 2022. Available at: <https://www.nasa.gov/missions/project-mercury/how-11-deaf-men-helped-shape-nasas-human-spaceflight-program/> (accessed 28.09.2025).
- [19] Howell, E. (2020). *New SpaceX spacesuits get five-star rating from NASA astronauts*. [online] Space.com. Available at: <https://www.space.com/spacex-spacesuits-five-star-astronaut-review.html> [Accessed 10 Sep. 2025].
- [20] NASA (2025). *Spacesuits - NASA*. [online] Nasa. Available at: <https://www.nasa.gov/humans-in-space/astronauts/spacesuits/>.
- [21] Pearson, E. (2024). *The astronaut wears Prada - why the fashion house is helping to design new spacesuits for moonwalkers*. [online] BBC Sky at Night Magazine. Available at: <https://www.skyatnightmagazine.com/news/prada-space-suits-axiom> [Accessed 28 Sep. 2025].
- [22] Diaz, J. (2024). *Why SpaceX suits look so sleek, and NASA's look so clunky*. [online] Fast Company. Available at: <https://www.fastcompany.com/91190436/why-spacex-suits-look-so-sleek-and-nasas-look-so-clunky> [Accessed 28 Sep. 2025].
- [23] Pearson, E. (2024b). *The astronaut wears Prada - why the fashion house is helping to design new spacesuits for moonwalkers*. [online] BBC Sky at Night Magazine. Available at: <https://www.skyatnightmagazine.com/news/prada-space-suits-axiom> [Accessed 28 Sep. 2025].
- [24] Chen, M., Goyal, R., Majji, M. and Skelton, R.E. (2021). Review of space habitat designs for long term space explorations. *Progress in Aerospace Sciences*, [online] 122, p.100692. doi:<https://doi.org/10.1016/j.paerosci.2020.100692>.
- [25] ESA – Parastronaut Feasibility Project Apply now to become an ESA astronaut – including the Parastronaut Feasibility Project (Press Release, 2021). https://www.esa.int/Newsroom/Press_Releases/Apply_now_to_become_an_ESA_astronaut
- [26] Robinson, D.K.R., Sterenborg, G., Häuplik, S. and Aguzzi, M. (2008). Exploring the challenges of habitation design for extended human presence beyond low-earth orbit: Are new requirements and processes needed? *Acta Astronautica*, 62(12), pp.721–732. doi:<https://doi.org/10.1016/j.actaastro.2008.01.034>.
- [27] NASA (2023). *SpaceX Spacesuit - NASA*. [online] NASA. Available at: https://www.nasa.gov/image-detail/amf-ksc-20180102-ph_spx01_0010/.
- [28] Jha, N. (2021). *Design Concepts in Human Space Missions*. [online] Available at: https://www.researchgate.net/publication/354177152_Design_Concepts_in_Human_Space_Missions.
- [29] WhereIsTheNorth 2024. (2025). *Universal Space Theory In Architecture*. [online] www.whereisthenorth.com. Available at: <https://www.whereisthenorth.com/article/universal-space-theory-in-architecture> [Accessed 28 Sep. 2025].
- [30] Cohen, M. (1998). Space Habitat Design Integration Issues. *International Conference on Evolvable Systems*, 981800. doi:<https://doi.org/10.4271/981800>.
- [31] Wieland, M., Feger, S., Machulla, T. and Schmidt, A. (2022). *Challenges and Opportunities for Inclusive Space Missions*. [online] arXiv.org. Available at: <https://arxiv.org/abs/2209.12576> [Accessed 28 Sep. 2025].
- [32] Chen, M., Goyal, R., Majji, M. and Skelton, R.E. (2021b). Review of space habitat designs for long term space explorations. *Progress in Aerospace Sciences*, [online] 122, p.100692. doi:<https://doi.org/10.1016/j.paerosci.2020.100692>.